

Tech Review Proposal on ColBERT Using the RAGatouille Library

Kaiyu Liu

kaiyul3@illinois.edu

University of Illinois Urbana-Champaign

Urbana, Illinois, USA

Yuyang Zeng

yuyang16@illinois.edu

University of Illinois Urbana-Champaign

Urbana, Illinois, USA

Abstract

This proposal outlines a technical review of the ColBERT retrieval model. We aim to utilize the RAGatouille library to demonstrate practical implementation and compare its performance against traditional sparse retrieval methods using the SciFact dataset.

Keywords

Information Retrieval, ColBERT, RAGatouille, Late Interaction, BERT

ACM Reference Format:

Kaiyu Liu and Yuyang Zeng. 2026. Tech Review Proposal on ColBERT Using the RAGatouille Library. In *Proceedings of Technical Review Project (CS410 Tech Review Proposal)*. ACM, New York, NY, USA, 1 page.

1 Summary

1.1 Technology under Review

ColBERT (Contextualized Late Interaction over BERT) is a retrieval model designed to make search both fast and accurate. ColBERT is primarily used as a First-Stage Retriever or a high-performance Reranker. Unlike traditional BERT-based rankers that must process every query-document pair together through a massive network, ColBERT independently encodes queries and documents into sets of token-level embeddings [1].

RAGatouille is an open-source Python library designed to bridge the gap between advanced Information Retrieval research and practical RAG (Retrieval-Augmented Generation) implementations [2]. It specifically focuses on making ColBERT as easy to use and deploy as standard dense embedding models.

1.2 Proposed Activities

In this project, we will conduct a reproduction study of the ColBERT retrieval architecture using the high-level RAGatouille library [2].

Our work will establish a baseline using the BM25 algorithm on the SciFact dataset from the BEIR benchmark [3]. This dataset consists of approximately 5,000 scientific claims paired with a corpus of 5,000 research abstracts, allowing us to quantify the performance gap between traditional sparse retrieval and modern neural approaches.

We will then implement the ColBERT pipeline, focusing on its unique "Late Interaction" mechanism [1]. A main objective of this tech review will be an ablation study on residual compression, a technique that reduces the index size by storing only the 'residual' difference between a vector and its nearest centroid. We also intend to experiment with different quantization levels to review the relationship between memory efficiency and retrieval effectiveness.

1.3 Complexity Estimation

The estimated time for this technical review is 30 hours, divided into four phases:

- (1) **Environment Setup:** Install and verify the CUDA/PyTorch stack needed for ColBERT, followed by a BM25 baseline execution on SciFact [3].
- (2) **Deployment & Ablation:** Deploy the ColBERT architecture via RAGatouille [2] to conduct an ablation study on residual compression. We will vary the `nbits` parameter to quantify the relationship between memory footprint and retrieval precision (MRR).
- (3) **Evaluation:** Utilize the `ranx` library to compute standardized metrics including NDCG@10, Mean Reciprocal Rank (MRR), and Mean Average Precision (MAP), followed by a Paired Student's t-test for statistical significance.
- (4) **Synthesis:** Synthesize findings into a formal technical review report.

This work is feasible within the requested timeframe, as RAGatouille library abstractions enable the efficient indexing of small-to-medium datasets like SciFact.

References

- [1] Omar Khattab and Matei Zaharia. 2020. ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '20)*. ACM, New York, NY, USA, 727–736.
- [2] Benjamin Clavié. 2023. RAGatouille: Making ColBERT accessible for everyone. GitHub repository. <https://github.com/bclavie/RAGatouille>
- [3] David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. 2020. Fact or Fiction: Verifying Scientific Claims. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. ACL, 7534–7550.